$=(u_1+u_2)+(v_2+v_1)$ (Assosciative Law)

$=(u_1+u_2)+(v_1+v_2)$ (Commutative Law)

Induction hypothesis: For any arbitrary $(n-1)$ terms of $(u_1, u_2, \ldots, u_{n-1})$ and $(v_1, v_2, \ldots, v_{n-1})$, $(u_1+v_1)+(u_2+v_2)+\ldots+(u_{n-1}+v_{n-1})$

$=(u_1+u_2+\ldots+u_{n-1})+(v_1+v_2+\ldots+v_{n-1})$

Induction Step: $(u_1+v_1)+(u_2+v_2)+(u_3+v_3)+\ldots+(u_n+v_n)$

$=(u_1+v_1)+((u_2+v_2)+(u_3+v_3)+\ldots+(u_n+v_n))$ (Associative Law)

$=(u_1+v_1)+((u_2+u_3+\ldots+u_n)+(v_2+v_3+\ldots+v_n))$ (By Induction Hyp)

$=((u_1+v_1)+(u_2+u_3+\ldots+u_n))+(v_2+v_3+\ldots+v_n)$ (Assosciative Law)

$=(u_1+(v_1+(u_2+u_3+\ldots+u_n)))+(v_2+v_3+\ldots+v_n)$ (Assosciative Law)

$=(u_1+((u_2+u_3+\ldots+u_n)+v_1))+(v_2+v_3+\ldots+v_n)$ (Commutative Law)

$=((u_1+(u_2+u_3+\ldots+u_n))+v_1)+(v_2+v_3+\ldots+v_n)$ (Assosciative Law)

$=(u_1+(u_2+u_3+\ldots+u_n))+(v_1+(v_2+v_3+\ldots+v_n))$ (Assosciative law)

$=(u_1+u_2+u_3+\ldots+u_n)+(v_1+v_2+v_3+\ldots+v_n)$ $\square$

From this proof, we can fill in the gaps in the proof of part (a).

Proof of Theorem 4

As was pointed out after Theorem 4, statements (a), (b) and (c) are logically equivalent. So, it suffices to show (for an arbitrary matrix A) that (a) and (d) are either both true or both false. This will tie all four statements together.

Let, (d) be True. $\therefore$ A has a pivot position in every row. By defn., a pivot position corresponds to the position of the leading 1 of a row in reduced row echelon form. $\therefore$ Every row of A has a leading 1, so there are no zero rows of A. So, in the corresponding augmented matrix of the eqn. $Ax=b$, there are no rows of the form $[0\ 0\ 0 \ldots 0\ b]$ where $b \neq 0$.